

ABHISHEK

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SUMMARY

Results-driven Data Scientist with 3+ years of experience in leveraging machine learning, deep learning, Generative AI and statistical methods to tackle high-impact challenges. Expertly developed and deployed predictive models, executed data cleaning and preprocessing tasks, and uncovered actionable insights from extensive datasets. Demonstrated strong analytical acumen and adaptability, consistently driving optimization and innovation in data-driven business solutions.

EXPERIENCE

DATA SCIENCE AND AI DEVELOPER – NOGIZ [MOHALI]

Sep 2023 – Present

- **Engineered a real-time multilingual voice tutor** using Whisper/Faster-Whisper, Silero VAD, RNNoise, Groq, VoxCPM, Qdrant, MongoDB and WebSockets, implementing streaming ASR/TTS, barge-in, RAG, zero-shot voice cloning, JWT authentication, multi-user sessions and per-user data isolation; introduced FIFO GPU scheduling and latency telemetry, achieving **400–600 ms end-to-end latency without RAG and 800–1,200 ms with RAG**.
- **Architected NEXUS, an adaptive pedagogy engine** combining Bayesian Knowledge Tracing (BKT), DINA, misconception tracking, affect modeling and SM-2 spaced repetition to dynamically sequence learning content based on learner state.
- **Built an autonomous AI newsroom** using LangGraph, Groq, SerpAPI, Google Trends, WordPress GraphQL and vector retrieval, orchestrating **10+ specialized agents** for trend discovery, research, content generation, review, SEO/AEO optimization and publishing; implemented fault-tolerant model fallback, duplicate-topic detection, semantic internal linking and editorial memory. [Link](#)
- **Engineered an autonomous job-application platform** using LangGraph, Playwright, FastAPI/Flask, Redis/RQ and SQLite, automating job discovery, JD extraction, browser-based application workflows and recruiter-question answering; developed DOM-based browser execution, persistent authentication state, resume-grounded responses and asynchronous task queues.
- **Architected a multi-tenant AWS serverless platform** using Lambda, API Gateway, DynamoDB and Cognito with MFA/OAuth2 authentication, event-driven tenant provisioning, JWT authorization, least-privilege IAM and tenant-level data isolation; deployed global web infrastructure using S3, CloudFront, Route53 and ACM and implemented **AWS CDK-based CI/CD**.
- **Developed enterprise RAG and agentic AI workflows** using Qdrant/vector stores, Sentence Transformers, Groq/Gemini and Azure AI Foundry, including an LMS assessment-planning agent and SQL-driven workflow agents; implemented schema-aware retrieval, adaptive template regeneration and automated quality-threshold validation.
- **Led development of ML forecasting and reinforcement-learning systems** using LTC/CfC/CNN models, Optuna and Ray RLlib; improved forecasting performance by **~12%** and simulated trading returns by **~15%**, with W&B experiment tracking and QuantConnect walk-forward backtesting.
- Implemented **adaptive template regeneration logic** to automatically revise assessment plans when early outputs failed quality thresholds. Built **agentic AI workflows** in Azure AI Foundry (NextGen): Planning agent-generated execution strategy, SQL agent synthesized and executed queries, Vector store provided schema-aware grounding with model context protocol.

Data Scientist Intern – INEURON.AI

JUN 2023 – Oct 2023

- Built an end-to-end credit card default prediction pipeline with automated data preprocessing, feature engineering, and validation workflows, reducing manual preparation time by **~40%**. Mitigated class imbalance using **SMOTE and undersampling**, and benchmarked multiple ML models with cross-validation and hyperparameter tuning, achieving **~90%** accuracy with SVM. Implemented **MLOps workflows** including model versioning, reproducible training pipelines, and CI/CD automation; deployed scalable real-time inference on **AWS** using SageMaker, Lambda, and API Gateway. Co-authored an **IEEE research paper** on MLOps-driven credit card default prediction, demonstrating scalable **pipeline** architecture and model lifecycle automation.

PROJECTS

BRIGHTSMILE – AI DENTAL APPOINTMENT BOOKING AGENT

[\(Link\)](#)

- **Architected and shipped** a production conversational AI agent that books dental appointments end-to-end, integrating a Groq LLM with a FastAPI + PostgreSQL backend to turn free-form patient messages into structured, validated bookings across a multi-stage collection flow. **Engineered** a resilient service-matching layer combining exact, synonym, and fuzzy (RapidFuzz) matching with a **deterministic fallback that keeps bookings flowing even when the LLM is unavailable**, plus intent detection to route conversations reliably. **Automated** a full CI/CD pipeline in GitHub Actions (pytest against a live Postgres service, Ruff, Bandit, Docker build) that deploys to Vercel serverless on a single push to main.
- **Hardened** the API for production with rate limiting, CORS, structured JSON logging, and health/readiness probes, then **deployed and diagnosed the live system** directly against Vercel's serverless runtime. **Owned the full lifecycle solo** — data modeling, LLM prompt design, deployment, and live production incident debugging — shipping a working customer-facing product under real-world constraints.

NEXUS – AI DENTAL APPOINTMENT BOOKING AGENT

- **Engineered** a low-latency, on-device voice assistant in Python (Flask, Socket.IO) achieving **400–600 ms speech-to-speech latency** by orchestrating streaming Faster-Whisper/CTranslate2 ASR, Silero VAD, Kokoro ONNX-Runtime TTS, and a Groq-hosted Llama 3.1 LLM with parallel streaming tool-calling across 18 tools. **Optimized** inference for constrained and variable GPU hardware through int8 Whisper quantization, CTranslate2-based CUDA detection, and automatic CPU fallback that keeps the ONNX-Runtime pipeline serving when CUDA/cuDNN libraries are unavailable.

MULTIMODAL RAG – WHISPER, CLIP, LLAVA AND FAISS

[\(GitHub Link\)](#)

- Created a **multimodal RAG** pipeline using **Whisper**, **CLIP**, and **FAISS** to process and index **video**, **audio**, and **text** data. Demonstrated expertise in **NLP**, **data retrieval**, and optimizing model performance for high-accuracy tasks.
- Improved **data retrieval accuracy** and **model inference** by processing speech, text, and image data in real-time.

RAG FROM SCRATCH – TECHNOLOGY LLAMA 3 (8B)

[\(GitHub Link\)](#)

- Developed a local **Retrieval-Augmented Generation (RAG)** pipeline utilizing **Llama 3**, enhancing **semantic search** and **data retrieval** for large documents. Optimized **model inference** on **consumer-grade hardware**, improving **latency** and **hardware utilization** for real-time use cases. Implemented graph theory algorithms for abuse prevention in customer lifecycle monitoring, enhancing data integrity.
- Integrated **prompt-engineering** systems to optimize responses and increase the quality of outputs from **large language models**.

CREDIT CARD DEFAULTS PREDICTIONS – TECHNOLOGY MLOPS

[\(GitHub Link\)](#)

- Designed an end-to-end **MLOps pipeline** on **AWS**, integrating data extraction, model training, and **real-time predictions**, maintaining, and monitoring the whole pipeline
- Addressed **class imbalances**, improved **model generalization**, and utilized **advanced feature engineering** to enhance prediction accuracy. Applied **performance testing** techniques to ensure model stability and reliability in production environments.

EDUCATION

Master of Science in Data Science – CHANDIGARH UNIVERSITY (MOHALI, PUNJAB)

2022 – 2024

Bachelor of Computer Application – B.B. S PUBLIC DEGREE COLLEGE (PRAYAGRAJ, UP)

2019 – 2022

OTHER

- **Languages:** English, Hindi
- **Soft Skills:** Problem-Solving, Communication, Task Management, **Customer Experience**, **Root Cause Analysis**, **Task Execution**
- **Certifications:** ML Specialization, Convolution Neural Network, Intro to ML in Production, Improving Neural Network, Python for Data Science.
- **Extra:** Presently learning diffusion models, and computer vision with generative AI
- **Published** a research-paper titled "Automated Credit Card Default Prediction Using MLOps on AWS: From Pipeline Development to Real-Time Deployment" [DOI: 10.1109/TIACOMP64125.2024.00050](https://doi.org/10.1109/TIACOMP64125.2024.00050)